Sample Exam

Medical Image Analysis

Student name:____

Immatriculation number:____________________________________

Exam Questions

Note: All the questions have equal weight.

1. In 2D acquisitions the slices are acquired one after the other in a pre-defined direction and stacked into a 3D volume; in axial acquisitions the slices lie in the transverse plane. Voxel sizes are given as in-plane (two values) and through-plane (one value, the slice thickness), and the field-of-view is image size $\times$ voxel size. A *dynamic* (4D) acquisition repeats the whole volume a number of times, one volume per time point.

   An axial dynamic acquisition is performed with in-plane voxel size 1.5×1.5 mm^2 and 128 pixels in both directions, through-plane voxel size 5.0 mm with 30 slices, and 200 time points acquired at a temporal resolution of 2.5 s per volume.

   (a) What is the field-of-view in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

   > **Solution:** Answer: $150 \times 192 \times 192$ mm^3.
   > The through-plane direction of an axial acquisition is head-to-toe: $30 \times 5.0 = 150$ mm. Both in-plane directions give $128 \times 1.5 = 192$ mm.

   (b) How many voxels does the complete 4D data set contain, and how long does the acquisition take?

   > **Solution:** Answer: 98 304 000 voxels, and 500 s = 8 min 20 s.
   > Per volume: $128 \times 128 \times 30 = 491\,520$ voxels. Over 200 time points: $491\,520 \times 200 = 98\,304\,000$. Duration: $200 \times 2.5 = 500$ s.

   (c) A second scan of the same patient is acquired at $1.0 \times 1.0 \times 1.0$ mm^3 isotropic. To compare measurements between the two scans, which property of the images has to be matched? Give the resize factors needed to bring one volume of the dynamic acquisition onto that grid, and the resulting image size.

   > **Solution:** Answer: the *voxel size* has to be matched (never the image size). Resize factors $5.0 \times 1.5 \times 1.5$ along head-to-toe, left-to-right and anterior-to-posterior, giving $150 \times 192 \times 192$ voxels.
   > The resize factor is the ratio of the old voxel size to the new one: $5.0/1.0 = 5.0$ head-to-toe and $1.5/1.0 = 1.5$ in the two in-plane directions. Applying these: $30 \cdot 5.0 = 150$ and

$128 \cdot 1.5 = 192$ twice. The field-of-view is unchanged, as it must be — only the sampling of it has changed.

2. Otsu's method selects a threshold for foreground–background separation automatically from the image histogram. Let p_1, p_2 be the numbers of pixels assigned to the two groups by a candidate threshold and σ_1^2, σ_2^2 the intensity variances within those groups. Which of the two criteria below is the one Otsu's method uses? Please circle the correct choice.

$$\min_{\tau} \quad p_1\sigma_1^2 + p_2\sigma_2^2 \qquad\qquad \max_{\tau} \quad p_1\sigma_1^2 + p_2\sigma_2^2$$

Solution: Answer: Left one — Otsu's method *minimises* the intra-class variance.

A good threshold is one for which the two resulting groups are each as homogeneous as possible, i.e. the spread of intensities *within* a group is small. Maximising the same quantity would deliberately choose the threshold that makes the two groups as internally inconsistent as possible, which is the worst threshold available.

3. Denoising can be written either as an energy minimisation or as a maximum-a-posteriori estimate. J is the observed noisy image and I the image we want to recover.

$$\max_{I} \ \log P(J \mid I) + \log P(I) \qquad\qquad \min_{I} \ \lambda\, D(I, J) + R(I)$$

(a) Please indicate which term corresponds to which term in the two formulations.

Solution: Answer:

$$\begin{aligned} \log P(J \mid I) &\iff \lambda\, D(I, J) && \text{(data consistency)} \\ \log P(I) &\iff R(I) && \text{(regularisation)} \end{aligned}$$

The likelihood encodes how the noise process works; the prior encodes which images are considered plausible before any data is seen.

(b) For total variation denoising the data term is $D(I, J) = \|I - J\|_2^2$ and the corresponding likelihood is Gaussian, $P(J \mid I) = \mathcal{N}(J;\, I,\, \sigma^2)$. What is λ in terms of σ, and what does that mean in words?

Solution: Answer: $\lambda = 1/(2\sigma^2)$.
Taking $-\log$ of a Gaussian likelihood gives $\|I - J\|_2^2/(2\sigma^2)$ up to a constant, so the weight in front of the data term *is* the inverse noise variance. In words: the weighting between data term and regulariser is not an arbitrary tuning knob — it states how much noise you believe is in the observation. A noisier image (large σ) means a small λ, which trusts the data less and lets the prior smooth more.

4. Please indicate whether the statement is true or false by circling the correct choice in each of the following. A denotes a binary image viewed as a set of pixels, B a structural element, and B_x the structural element translated to x.
   (a) **TRUE** False The dilation of A by B is the set of all x for which B_x has a non-empty intersection with A.
   (b) True **FALSE** The opening of A by B is a dilation followed by an erosion, and it fills small gaps and holes.
   (c) True **FALSE** Morphological operations are linear and shift-invariant.
   (d) **TRUE** False Applying a median filter to a label map cannot introduce labels that were not already present in it.

> **Solution:** (a) TRUE — $A \oplus B \triangleq \{x \mid B_x \cap A \neq \emptyset\}$. Erosion is the stricter condition $A \ominus B \triangleq \{x \mid B_x \subseteq A\}$.
>
> (b) FALSE — both halves are wrong, and they are wrong together. *Opening* is an erosion followed by a dilation, $(A \ominus B) \oplus B$, and it removes small islands and protrusions. *Closing* is the other order, $(A \oplus B) \ominus B$, and that is the one that fills small gaps and holes.
>
> (c) FALSE — they are shift-invariant, but they are emphatically *non-linear*. That is precisely why they can remove an isolated island without blurring the boundary of the structure next to it, which no linear filter can do.
>
> (d) TRUE — the median of a set of values is always one of those values, so no new value can appear. This is what makes median filtering a safe post-processing step on a segmentation: it removes isolated mislabelled voxels without inventing a label that does not exist. Averaging filters do not have this property.

5. In landmark-based registration a set of corresponding landmark pairs is used to determine the transformation parameters. How many corresponding landmark pairs are needed for the system to be exactly determined when the transformation model is (i) a 2D affine transformation, and (ii) a 3D affine transformation? Explain the counting.

> **Solution:** Answer: (i) **3** landmark pairs in 2D, (ii) **4** landmark pairs in 3D.
>
> A 2D affine transformation has 6 free parameters ($A \in \mathbb{R}^{2\times 2}$ plus $t \in \mathbb{R}^2$). Each landmark correspondence must match both coordinates of the point and therefore contributes 2 scalar equations. Setting $2N = 6$ gives $N = 3$.
>
> A 3D affine transformation has 12 free parameters ($A \in \mathbb{R}^{3\times 3}$ plus $t \in \mathbb{R}^3$), and each correspondence contributes 3 equations, so $3N = 12$ gives $N = 4$.
>
> With more landmarks than this the system becomes over-determined and is solved in the least-squares sense: the landmarks themselves are then no longer aligned exactly, but the rest of the image is aligned better — which is usually what you actually want.

6. Optimisation in image registration is difficult, and the objective function generally has local minima that a gradient-based optimiser can get stuck in. Please describe the two practical strategies presented in the lecture for dealing with this, and briefly explain why each of them helps. You do not need to write down the equations.

Solution: Answer: *Sequential optimisation.* First align the images with a **linear** transformation model (rigid or affine), and only then run the deformable, non-linear registration. The linear model has very few parameters and removes the large global misalignment, so the non-linear stage starts close to the correct answer and only has to explain the local differences that remain. Running the deformable model straight away would ask it to absorb a global rotation or translation through local deformation, which is both harder to optimise and physically wrong.

Multi-scale optimisation. Build an image pyramid and start the registration at the **coarsest** scale, using the result of each scale to initialise the next finer one, $\theta_0^{(k-1)} = \theta_*^{(k)}$. A coarse image has fewer details, so its objective function is smoother and has fewer local minima; the coarse solution captures the large-scale alignment and places the finer optimisation inside the basin of the correct minimum.

The two are routinely combined, and both are really the same idea: solve an easier version of the problem first and use its answer as the starting point for the harder one.

7. Convolutional networks reduce the spatial dimensions of their feature maps either with pooling or with strided convolutions. Which of the following statements is **inaccurate**?
   - ○ (A) Max pooling provides partial local translation invariance: the position of the maximum can move within the pooling window without changing the pooled value.
   - ○ (B) Pooling is applied to each channel separately and has no learnable parameters at all.
   - ○ (C) A strided convolution reduces the spatial dimensions and, like pooling, buys a degree of translation invariance.
   - ○ (D) Average pooling is a linear operation, whereas max pooling is not.

Solution: Answer: C.

Strides help with dimensionality reduction, and that is all they do. Increasing the stride simply samples the convolution at fewer positions — it discards information (the more so the larger the stride) and gives no invariance whatsoever, because shifting the input by one pixel changes which positions get sampled. Max pooling is different precisely because of the maximum: the largest activation inside the window can move around inside that window and the output does not change. That is a real, if partial and purely local, invariance.

8. Three standard measures are used to judge how well a statistical shape model has embedded the training data into a lower-dimensional space. Which of the following statements is **incorrect**?
   - ○ (A) Compactness measures how much of the total variance is explained by the first t modes, $C(t) = \sum_{i \le t} \lambda_i / \lambda_{total}$; a compact model needs few parameters to generate new shape instances.
   - ○ (B) Generalisation measures the ability of the model to represent instances that were not in the training set, and is estimated with leave-one-out reconstruction.
   - ○ (C) Specificity measures how accurately the model reconstructs the training shapes themselves; a perfectly specific model reproduces every training instance exactly.

○ (D) Principal component analysis assumes that the data follow a Gaussian distribution, and produces an orthogonal lower-dimensional coordinate system that maximises the variance along each successive axis.

Solution: Answer: C.

Specificity asks the opposite question: does the model generate *only valid* shapes? It is measured by drawing a large number of random instances from the model and, for each, computing the distance to the closest shape in the training set. A model that can generate anatomically impossible shapes scores badly, however well it reconstructs the training data.

Note the tension this creates with (B): generalisation improves as you keep more modes, specificity degrades. The number of modes retained is a trade-off between the two.

9. Which of the following statements about positional encoding and the structure of a transformer layer is **incorrect**?

○ (A) Without positional encoding a transformer is equivariant to permutations of its input tokens, so "I bought an apple watch" and "watch an apple I bought" cannot be distinguished.

○ (B) An ideal positional encoding should give a unique representation for each position, be bounded, generalise to sequences longer than those seen in training, and encode the relative positions of tokens.

○ (C) In the sinusoidal scheme the positional vector is concatenated to the token embedding, which doubles the dimension of the representation entering the first transformer layer.

○ (D) A transformer layer consists of multi-head self-attention followed by a per-token MLP, each of the two wrapped in a residual connection and followed by layer normalisation.

Solution: Answer: C.

The positional vector is **added**, not concatenated: $\tilde{x}_i = x_i + r_i$ with $r_i, x_i \in \mathbb{R}^{D\times 1}$. The dimension of the representation is therefore unchanged. This is why the encoding has to be bounded — an unbounded positional vector added to the embedding would swamp the token's own content.

10. The magnitude image of an MR acquisition is formed from the real and imaginary channels as $J(x) = \sqrt{(I_r(x)+\epsilon)^2 + (I_i(x)+\eta)^2}$, where $\epsilon, \eta \sim \mathcal{N}(0,\sigma)$ are the noise in the two channels. What kind of noise does the resulting magnitude image have, and why does it matter?

○ (A) Additive Gaussian noise, since ϵ and η are both Gaussian and the operations applied to them are continuous.

○ (B) Multiplicative speckle noise, of the same type as in ultrasound and optical coherence tomography.

○ (C) Rician noise — taking the magnitude makes the noise non-Gaussian and signal-dependent, so denoising methods derived from an additive Gaussian model are mis-specified, particularly in low-signal regions.

○ (D) Poisson noise, of the same type as the quantum noise in low-dose CT.

◯ (E) No noise at all, because the two independent noise terms cancel in the magnitude operation.

Solution: Answer: C.

The noise is Gaussian in the real and imaginary channels, but the magnitude operation is non-linear, and a non-linear function of Gaussian variables is not Gaussian. The result is the **Rician** distribution. The practical consequence is that the noise is signal-dependent: where the true signal is strong the Rician distribution is close to Gaussian and an additive model is a fair approximation, but in dark, low-signal regions it is strongly skewed and even biases the mean intensity upwards. This is why specialised Rician denoisers exist for MRI.

11. Consider a classification network that takes grey-scale input images of size 60×60 with one input channel. All convolutions are *valid* (no padding, $P = 0$) with stride $S = 1$, and each is followed by max pooling with $K = 2 \times 2$ and $S = 2$:

- **Conv1**: 12 kernels with $K = 5 \times 5$ **Pool1**: 2×2, $S = 2$
- **Conv2**: 24 kernels with $K = 5 \times 5$ **Pool2**: 2×2, $S = 2$
- **Conv3**: 48 kernels with $K = 3 \times 3$ **Pool3**: 2×2, $S = 2$

Recall that a valid convolution gives $N_l = N_{l-1} - k_1 + 1$. Circle True or False:

(a) **TRUE** False The output of Pool3 has a spatial size of 5×5.

(b) True **FALSE** Each neuron in Pool3 has a global receptive field covering the entire 60×60 input image.

(c) **TRUE** False The total number of convolutional kernel weights (excluding the biases) is $5 \cdot 5 \cdot 1 \cdot 12 + 5 \cdot 5 \cdot 12 \cdot 24 + 3 \cdot 3 \cdot 24 \cdot 48 = 17\,868$.

(d) True **FALSE** Replacing the three max-pooling layers by 2×2 convolutions with stride 2 would leave the total number of learnable parameters unchanged, because pooling and strided convolution both simply downsample.

Solution: (a) TRUE — tracking the sizes: $60 \xrightarrow{\text{Conv1}} 56 \xrightarrow{\text{Pool1}} 28 \xrightarrow{\text{Conv2}} 24 \xrightarrow{\text{Pool2}} 12 \xrightarrow{\text{Conv3}} 10 \xrightarrow{\text{Pool3}} 5$.

(b) FALSE — it is 28×28, less than a quarter of the image area. Track the receptive field R and jump j from $R = 1,\ j = 1$ with $R \leftarrow R + (K - 1) \cdot j$ and $j \leftarrow j \cdot S$: Conv1 $\rightarrow R = 5,\ j = 1$; Pool1 $\rightarrow R = 6,\ j = 2$; Conv2 $\rightarrow R = 14,\ j = 2$; Pool2 $\rightarrow R = 16,\ j = 4$; Conv3 $\rightarrow R = 24,\ j = 4$; Pool3 $\rightarrow R = 28,\ j = 8$. Output size and receptive field are different things — a 5×5 output does not mean each of those 25 neurons sees everything. (The global stride, incidentally, is 8×8.)

(c) TRUE — $300 + 7\,200 + 10\,368 = 17\,868$. Pooling layers contribute nothing.

(d) FALSE — max pooling has *no* learnable parameters, whereas convolutions do. The replacement would add $2 \cdot 2 \cdot 12 \cdot 12 + 2 \cdot 2 \cdot 24 \cdot 24 + 2 \cdot 2 \cdot 48 \cdot 48 = 576 + 2\,304 + 9\,216 = 12\,096$ new weights, an increase of about 68%. The two operations also differ in behaviour, not just in cost: pooling buys partial local translation invariance, strided convolution does not.

12. Which of the following statements on domain shift, adaptation and uncertainty are correct (multiple answers are possible)?

○ (A) Acquisition shift is a change in $P_D(X \mid Z)$ — scanner, resolution, contrast, protocol — and is the kind of dataset shift encountered most often in medical imaging.

○ (B) Prevalence shift is a change in $P_D(Y \mid X)$, caused for instance by differences in annotation policy or in the experience of the annotator.

○ (C) The Expected Calibration Error measures the average gap between the confidence a model assigns to its predictions and the accuracy it actually achieves at that confidence.

○ (D) Adapting only the batch-normalisation parameters of a network is not possible, because batch normalisation has no learnable parameters.

○ (E) A deep ensemble approximates $p(\theta \mid D, M)$ by training the same architecture several times from different random initialisations, on the assumption that the different runs land in different modes of the parameter posterior.

Solution: Answer: (A), (C) and (E) are correct.

(B) confuses two rows of the taxonomy. *Prevalence* shift is a change in $P_D(Y)$ — the case–control balance, or how the target population was selected. A change in $P_D(Y \mid X)$ driven by annotation policy or annotator experience is *annotation* shift.

(D) is wrong: batch normalisation computes $\mathrm{BN}(a_l) = \gamma\,(a_l - \mu_l)/\sqrt{\sigma_l^2 + \epsilon} + \beta$, and the scale γ and shift β are learnable. Adapting only those is a standard lightweight strategy, attractive because it touches very few parameters and leaves the expensively trained weights alone — though it is not obvious that they are the right parameters for correcting a change in intensity characteristics.